

# Engineering Mathematics

A compact learning, formula, practice and examination handbook for Diploma Curriculum Revision 2026 Course 1002.

## POLY PMNA

|                 |         |
|-----------------|---------|
| Course Code     | 1002    |
| Semester        | 1       |
| Credits         | 4       |
| Contact Hours   | 60      |
| Teaching Scheme | 3:1:0:0 |
| CIE             | 40      |
| ESE             | 60      |
| Bloom Level     | Apply   |

**Core areas:** Matrices and Determinants, Trigonometry, Coordinate Geometry and Differential Calculus.

### COURSE OVERVIEW

## What Course 1002 expects

Engineering Mathematics is not a proof-heavy university text. The target is correct selection and application of standard methods.

### Course purpose

The course supplies the mathematical tools required to represent quantities, solve simultaneous relations, analyse angles and geometry, and calculate limits and rates of change in engineering situations.

## Prerequisites

- Linear and quadratic equations
- Rectangular coordinates
- Angles and basic arithmetic
- Secondary-education-level algebra

## Teaching and assessment

**60 contact hours** with L:T:P:R = 3:1:0:0. Continuous Internal Evaluation carries 40 marks and the End Semester Examination carries 60 marks.

## Learning approach

Read the idea, note the formula, follow the method, solve the worked example, then complete the short practice without looking at the answer.

## Course outcomes

| CO  | Outcome                                                                                                | Module | Bloom level |
|-----|--------------------------------------------------------------------------------------------------------|--------|-------------|
| CO1 | Apply matrices and determinants to solve systems of linear equations involving two and three unknowns. | I      | Apply       |
| CO2 | Use trigonometric identities and functions to solve trigonometric problems.                            | II     | Apply       |
| CO3 | Interpret problems involving straight lines and circles using coordinate geometry.                     | III    | Apply       |
| CO4 | Evaluate limits and derivatives.                                                                       | IV     | Apply       |

## Module I

**11 L + 4 T**

Matrices, determinants, inverse and Cramer's rule.

## Module II

**11 L + 4 T**

Angles, ratios, ASTC, reduction and compound angles.

## Module III

**11 L + 4 T**

Straight lines and circles.

## Module IV

**12 L + 3 T**

Limits, derivatives, rules and second derivatives.

**Exam structure:** each module contributes 15 marks: 2 one-mark questions, 6 marks from Part B, and one 7-mark choice set.

## STUDY METHOD

### How to use this handbook

#### 1. Learn

Read one topic at a time. Copy the formula and understand what each symbol means.

## 2. Practise

Cover the solution and repeat the numbered steps. Do not jump directly to the final answer.

## 3. Check

Verify signs, substitution, units and whether the final result satisfies the original condition.

**Bad study method:** memorising an answer pattern without checking which rule is required. Similar-looking questions may need different methods.

### QUICK REFERENCE

## Formula and rule bank

### Matrices

**2 × 2 determinant:**

$$|A| = ad - bc$$

**Inverse:**

$$A^{-1} = (1/|A|) \text{ adj } A, |A| \neq 0$$

**Cramer:**  $x = \Delta_x/\Delta$ ,  $y = \Delta_y/\Delta$ ,  $z = \Delta_z/\Delta$ .

### Trigonometry

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \text{cosec}^2\theta$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

## Straight lines

$$m = (y_2 - y_1) / (x_2 - x_1)$$

$$m = \tan \theta$$

$$y - y_1 = m(x - x_1)$$

$$\tan \theta = |(m_2 - m_1) / (1 + m_1 m_2)|$$

## Circles

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{Centre} = (-g, -f)$$

$$r = \sqrt{g^2 + f^2 - c}$$

## Limits

Direct substitution when denominator is non-zero.

Factorise and cancel a common factor for 0/0 forms.

$$\lim_{x \rightarrow a} (x^n - a^n) / (x - a) = na^{n-1}$$

$$\lim_{x \rightarrow 0} \sin x / x = 1, x \text{ in radians.}$$

## Derivatives

$$d(k)/dx = 0; d(x^n)/dx = nx^{n-1}$$

$$d(e^x)/dx = e^x; d(\log x)/dx = 1/x$$

$$(uv)' = u'v + uv'$$

$$(u/v)' = (vu' - uv')/v^2$$

**Malayalam support:** Formula മാത്രം memorise ചെയ്യരുത്; question-ൽ നൽകിയ form തിരിച്ചറിഞ്ഞ് ശരിയായ rule തിരഞ്ഞെടുക്കുക.

MODULE I

Matrices and Determinants

Represent data in rectangular form, perform matrix operations, evaluate determinants and solve simultaneous equations.

11 Lecture Hours

4 Tutorial Hours

CO1

## 1.1 Matrix, order and special types–

A **matrix** is a rectangular arrangement of numbers in rows and columns. If a matrix has  $m$  rows and  $n$  columns, its order is  $\mathbf{m} \times \mathbf{n}$ . The element in row  $i$ , column  $j$  is written  $a_{ij}$ .

| Type             | Recognition rule                             | Example                                                             |
|------------------|----------------------------------------------|---------------------------------------------------------------------|
| Row matrix       | One row                                      | $[2 \quad -1 \quad 5]$                                              |
| Column matrix    | One column                                   | $\begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix}$                        |
| Square matrix    | Rows = columns                               | $2 \times 2$ or $3 \times 3$                                        |
| Diagonal matrix  | All off-diagonal elements are zero           | $\text{diag}(4, -2, 7)$                                             |
| Upper triangular | All entries below the main diagonal are zero | $\begin{bmatrix} 2 & 1 & 3 \\ 0 & 5 & 4 \\ 0 & 0 & 6 \end{bmatrix}$ |
| Lower triangular | All entries above the main diagonal are zero | $\begin{bmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 3 & 4 & 6 \end{bmatrix}$ |
| Identity matrix  | Diagonal entries 1; other entries 0          | $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                |
| Zero matrix      | Every entry is zero                          | $O_{2 \times 3}$                                                    |

**Order first:** two matrices can be equal, added or subtracted only when their orders are the same.

## 1.2 Transpose and algebra of matrices—

The transpose  $A^T$  is formed by changing rows into columns. If  $A$  is  $m \times n$ , then  $A^T$  is  $n \times m$ .

If  $A = [[a,b,c],[d,e,f]]$ , then  $A^T = [[a,d],[b,e],[c,f]]$ .

| Operation              | Condition / rule                            | Result order                              |
|------------------------|---------------------------------------------|-------------------------------------------|
| Scalar multiplication  | Multiply every element by the scalar $k$    | Same as $A$                               |
| Equality               | Same order and corresponding elements equal | Statement $A = B$                         |
| Addition / subtraction | Same order; operate corresponding elements  | Same as $A$ and $B$                       |
| $AB$                   | Columns of $A$ = rows of $B$                | (rows of $A$ ) $\times$ (columns of $B$ ) |

**Common mistake:** matrix multiplication is not element-by-element multiplication. Use row-by-column products. In general,  $AB \neq BA$ .



## 1.3 Determinants, minors and cofactors–

A determinant is a number associated with a square matrix. For  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ ,  $|A| = ad - bc$ .

For a  $3 \times 3$  determinant expanded along row 1:

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}.$$

The **minor**  $M_{ij}$  is the determinant remaining after deleting row  $i$  and column  $j$ .

The **cofactor** is  $C_{ij} = (-1)^{i+j}M_{ij}$ .

### Cofactor sign pattern

|   |   |   |
|---|---|---|
| + | – | + |
| – | + | – |
| + | – | + |

Expansion may be taken along any row or column. Choose one containing zeros when possible.

## 1.4 Adjoint and inverse of a $2 \times 2$ matrix—

For  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , interchange the main diagonal elements and change the signs of the off-diagonal elements.

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = 1/(ad-bc) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \text{ provided } ad-bc \neq 0.$$

Determinant zero ആണെങ്കിൽ matrix-ന് inverse ഉണ്ടായിരിക്കില്ല.

**Check the determinant before dividing.** Writing an inverse when  $|A| = 0$  is mathematically invalid.

## 1.5 Cramer's rule for two and three unknowns—

Cramer's rule applies to a square system when the coefficient determinant  $\Delta$  is non-zero.

- 1 Write the coefficient determinant  $\Delta$ .
- 2 Replace one coefficient column at a time by the constant column to obtain  $\Delta$

$x$  ,  $\Delta$

$y$  and, where required,  $\Delta$

$z$  .

- 3 Calculate  $x = \Delta$

$x$  / $\Delta$ ,  $y = \Delta$

$y$  / $\Delta$  and  $z = \Delta$

$z$  / $\Delta$ .

- 4 Substitute the answers in the original equations to check.

**Engineering mini-application:** simultaneous equations occur in current balance, force balance, production mixing and calibration problems. Cramer's rule converts the coefficient data into determinant calculations.

## Worked examples

## Solved Example 1.1: Order, type and transpose

**Question.** For  $A = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 5 \end{bmatrix}$ , state the order and find  $A^T$ .

**Rule.** Rows become columns.

- 1 A has 2 rows and 3 columns, so its order is  $2 \times 3$ .
- 2 The first row  $[3, -1, 2]$  becomes the first column.
- 3 The second row  $[0, 4, 5]$  becomes the second column.

**Final answer:**  $A^T = \begin{bmatrix} 3 & 0 \\ -1 & 4 \\ 2 & 5 \end{bmatrix}$ .

**Check:** The order changes from  $2 \times 3$  to  $3 \times 2$ .

## Solved Example 1.2: Addition and multiplication

**Question.** Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}$ . Find  $A+B$  and  $AB$ .

**Rule.** For  $AB$ , multiply each row of  $A$  by each column of  $B$ .

- 1  $A+B = \begin{bmatrix} 1+2 & 2+0 \\ 3-1 & 4+5 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 9 \end{bmatrix}$ .
- 2  $(AB)_{11} = 1(2) + 2(-1) = 0$ ;  
 $(AB)_{12} = 1(0) + 2(5) = 10$ .
- 3  $(AB)_{21} = 3(2) + 4(-1) = 2$ ;  
 $(AB)_{22} = 3(0) + 4(5) = 20$ .

**Final answer:**  $A+B = \begin{bmatrix} 3 & 2 \\ 2 & 9 \end{bmatrix}$  and  $AB = \begin{bmatrix} 0 & 10 \\ 2 & 20 \end{bmatrix}$ .

**Check:**  $BA$  is different, which confirms multiplication is order-sensitive.

### Solved Example 1.3: Determinant, minor and cofactor

**Question.** Evaluate  $|A|$  for  $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$ . Also find  $M_{12}$  and  $C_{12}$ .

**Rule.** Expand along the first row;  $C_{12} = -M_{12}$ .

- 1  $|A| = 1(4 \cdot 6 - 5 \cdot 0) - 2(0 \cdot 6 - 5 \cdot 1) + 3(0 \cdot 0 - 4 \cdot 1)$ .
- 2  $|A| = 24 - 2(-5) + 3(-4) = 24 + 10 - 12 = 22$ .
- 3 Delete row 1 and column 2:  $M_{12} = \begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} = 0 - 5 = -5$ .
- 4  $C_{12} = -M_{12} = 5$ .

**Final answer:**  $|A| = 22$ ,  $M_{12} = -5$  and  $C_{12} = 5$ .

### Solved Example 1.4: Inverse and a two-variable system

**Question.** Find  $A^{-1}$  for  $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ , then solve  $2x + y = 7$  and  $5x + 3y = 19$  by Cramer's rule.

**Rule.**  $A^{-1} = (1/|A|)\text{adj } A$ ;  $x = \Delta x / \Delta$  and  $y = \Delta y / \Delta$ .

- 1  $|A| = 2 \cdot 3 - 1 \cdot 5 = 1$ , so  $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ .
- 2  $\Delta = \begin{vmatrix} 2 & 1 \\ 5 & 3 \end{vmatrix} = 1$ .
- 3  $\Delta x = \begin{vmatrix} 7 & 1 \\ 19 & 3 \end{vmatrix} = 21 - 19 = 2$ .
- 4  $\Delta y = \begin{vmatrix} 2 & 7 \\ 5 & 19 \end{vmatrix} = 38 - 35 = 3$ .

**Final answer:**  $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ ,  $x = 2$  and  $y = 3$ .

**Check:**  $2(2) + 3 = 7$  and  $5(2) + 3(3) = 19$ .

### Solved Example 1.5: Three-variable Cramer's rule

**Question.** Solve  $x+y+z=6$ ,  
 $2x-y+z=3$ , and  $x+2y-z=2$ .

**Rule.** Calculate  $\Delta$ , then replace each coefficient column by  $[6,3,2]^T$ .

- 1  $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 7$ .
- 2  $\Delta x = \begin{vmatrix} 6 & 1 & 1 \\ 3 & -1 & 1 \\ 2 & 2 & -1 \end{vmatrix} = 7$ .
- 3  $\Delta y = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 14$ .
- 4  $\Delta z = \begin{vmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 1 & 2 & 2 \end{vmatrix} = 21$ .
- 5  $x=7/7=1$ ,  $y=14/7=2$ ,  $z=21/7=3$ .

**Final answer:**  $x=1$ ,  $y=2$ ,  $z=3$ .

**Check:** All three original equations are satisfied.

## Practice questions

State the order and type of  $[[4, -2, 1]]$ .

Find the transpose of  $[[1, 0], [3, 5], [-2, 4]]$ .

For  $A = [[2, 1], [-1, 3]]$  and  $B = [[4, 0], [2, -2]]$ , find  $2A - B$ .

Find  $AB$  for  $A = [[1, 2, 0], [-1, 3, 1]]$  and  $B = [[2, 1], [0, 4], [5, -2]]$ .

Evaluate  $\begin{vmatrix} 3 & 2 \\ 5 & 4 \end{vmatrix}$ .

Find the determinant of  $[[2, 0, 1], [3, -1, 2], [1, 4, 0]]$ .

Find the inverse of  $[[4, 1], [2, 1]]$ .

Solve  $3x+2y=12$ ,  $x-y=1$  by Cramer's rule.

Solve  $x+y+z=9$ ,  $2x-y+z=4$ ,  
 $x+2y-z=3$  by Cramer's rule.

1. Order  $1 \times 3$ ; row matrix.

2.  $[[1, 3, -2], [0, 5, 4]]$ .

3.  $2A - B = [[0, 2], [-4, 8]]$ .

4.  $AB = [[2, 9], [3, 9]]$ .

5. 2.

6.  $-3$ .

7.  $A^{-1} = (1/2)[[1, -1], [-2, 4]]$ .

8.  $x = 14/5$ ,  $y = 9/5$ .

9.  $x = 9/7$ ,  $y = 22/7$ ,  $z = 32/7$ .

**Module 1 common mistakes:** wrong matrix order, using element-wise multiplication, missing the negative cofactor sign, and using Cramer's rule when  $\Delta = 0$ .

## MODULE II

# Trigonometry

Identify angles, evaluate ratios, apply quadrant signs and use reduction and compound-angle formulae.

11 Lecture Hours

4 Tutorial Hours

CO2

## 2.1 Angles and trigonometric ratios—

An angle is formed by rotating a ray about its endpoint. Counter-clockwise rotation is positive and clockwise rotation is negative.

| Angle type | Range / measure                  |
|------------|----------------------------------|
| Acute      | $0^\circ < \theta < 90^\circ$    |
| Right      | $90^\circ$                       |
| Obtuse     | $90^\circ < \theta < 180^\circ$  |
| Straight   | $180^\circ$                      |
| Reflex     | $180^\circ < \theta < 360^\circ$ |
| Complete   | $360^\circ$                      |

$\sin \theta = \text{opposite/hypotenuse}$ ;  $\cos \theta = \text{adjacent/hypotenuse}$ ;  $\tan \theta = \text{opposite/adjacent}$ .  
 $\text{cosec } \theta = 1/\sin \theta$ ;  $\sec \theta = 1/\cos \theta$ ;  $\cot \theta = 1/\tan \theta$ .



## 2.2 Standard angles and Pythagorean relations—

| $\theta$                        | $0^\circ$ | $30^\circ$   | $45^\circ$   | $60^\circ$   | $90^\circ$  |
|---------------------------------|-----------|--------------|--------------|--------------|-------------|
| <b><math>\sin \theta</math></b> | 0         | $1/2$        | $1/\sqrt{2}$ | $\sqrt{3}/2$ | 1           |
| <b><math>\cos \theta</math></b> | 1         | $\sqrt{3}/2$ | $1/\sqrt{2}$ | $1/2$        | 0           |
| <b><math>\tan \theta</math></b> | 0         | $1/\sqrt{3}$ | 1            | $\sqrt{3}$   | Not defined |

$$\sin^2\theta + \cos^2\theta = 1$$

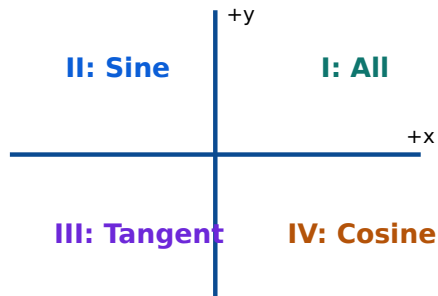
$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

The reciprocal ratios are not independent:  $\operatorname{cosec}=1/\sin$ ,  $\sec=1/\cos$  and  $\cot=1/\tan$ .

## 2.3 Ratios of any angle, ASTC and reduction formulae–

Use the reference angle and the sign in the required quadrant. The ASTC rule gives which basic ratios are positive.



ASTC: All, Sine, Tangent, Cosine are positive in quadrants I, II, III and IV respectively.

| Angle form           | sin            | cos            | tan            |
|----------------------|----------------|----------------|----------------|
| $90^\circ - \theta$  | $\cos \theta$  | $\sin \theta$  | $\cot \theta$  |
| $90^\circ + \theta$  | $\cos \theta$  | $-\sin \theta$ | $-\cot \theta$ |
| $180^\circ - \theta$ | $\sin \theta$  | $-\cos \theta$ | $-\tan \theta$ |
| $180^\circ + \theta$ | $-\sin \theta$ | $-\cos \theta$ | $\tan \theta$  |
| $360^\circ - \theta$ | $-\sin \theta$ | $\cos \theta$  | $-\tan \theta$ |

Quadrant കണ്ടെത്തിയ ശേഷം sign തീരുമാനിക്കുക; reference angle value പിന്നീട് ഉപയോഗിക്കുക.

## 2.4 Compound-angle formulae–

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$$

$$\tan(A-B) = (\tan A - \tan B) / (1 + \tan A \tan B)$$

**Sign trap:** cosine has the opposite sign pattern from sine. For  $\cos(A+B)$  use minus; for  $\cos(A-B)$  use plus.

**Engineering mini-application:** trigonometric ratios resolve an inclined force or displacement into horizontal and vertical components and help calculate slopes and orientations.

## Worked examples

### Solved Example 2.1: Standard-angle value

**Question.** Evaluate  $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ .

**Rule.** Recognise  $\sin(A+B)$ .

$$\text{1 } \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \sin(60^\circ + 30^\circ).$$

$$\text{2 } \sin 90^\circ = 1.$$

**Final answer: 1.**

### Solved Example 2.2: One acute ratio given

**Question.** If  $\tan \theta = 3/4$  and  $\theta$  is acute, find  $\sin \theta$  and  $\cos \theta$ .

**Rule.** Use a right triangle with opposite 3 and adjacent 4.

$$\text{1 } \text{Let opposite}=3 \text{ and adjacent}=4.$$

$$\text{2 } \text{Hypotenuse} = \sqrt{(3^2 + 4^2)} = 5.$$

$$\text{3 } \sin \theta = 3/5 \text{ and } \cos \theta = 4/5.$$

**Final answer:  $\sin \theta = 3/5$ ,  $\cos \theta = 4/5$ .**

### Solved Example 2.3: ASTC and reduction

**Question.** Find  $\sin 150^\circ$ ,  $\cos 210^\circ$  and  $\tan 330^\circ$ .

**Rule.** Use reference angles and quadrant signs.

- 1  $150^\circ = 180^\circ - 30^\circ$  lies in quadrant II:  $\sin$  positive, so  $\sin 150^\circ = \sin 30^\circ = 1/2$ .
- 2  $210^\circ = 180^\circ + 30^\circ$  lies in quadrant III:  $\cos$  negative, so  $\cos 210^\circ = -\cos 30^\circ = -\sqrt{3}/2$ .
- 3  $330^\circ = 360^\circ - 30^\circ$  lies in quadrant IV:  $\tan$  negative, so  $\tan 330^\circ = -\tan 30^\circ = -1/\sqrt{3}$ .

**Final answer:**  $1/2$ ,  $-\sqrt{3}/2$  and  $-1/\sqrt{3}$ .

### Solved Example 2.5: Tangent difference

**Question.** Evaluate  $\tan 15^\circ$  exactly.

**Rule.**  $15^\circ = 45^\circ - 30^\circ$ .

- 1  $\tan 15^\circ = (\tan 45^\circ - \tan 30^\circ) / (1 + \tan 45^\circ \tan 30^\circ)$ .
- 2  $= (1 - 1/\sqrt{3}) / (1 + 1/\sqrt{3})$ .
- 3 Multiply numerator and denominator by  $\sqrt{3}$ :  $(\sqrt{3} - 1) / (\sqrt{3} + 1)$ .
- 4 Rationalise to obtain  $2 - \sqrt{3}$ .

**Final answer:**  $\tan 15^\circ = 2 - \sqrt{3}$ .

### Solved Example 2.4: Exact compound angle

**Question.** Find  $\cos 75^\circ$  exactly.

**Rule.**  $75^\circ = 45^\circ + 30^\circ$  and  $\cos(A+B) = \cos A \cos B - \sin A \sin B$ .

- 1  $\cos 75^\circ = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$ .
- 2  $= (1/\sqrt{2})(\sqrt{3}/2) - (1/\sqrt{2})(1/2)$ .
- 3  $= (\sqrt{3} - 1) / (2\sqrt{2}) = (\sqrt{6} - \sqrt{2}) / 4$ .

**Final answer:**  $\cos 75^\circ = (\sqrt{6} - \sqrt{2}) / 4$ .

## Practice questions

Course 1002

Classify  $45^\circ$ ,  $90^\circ$ ,  $125^\circ$ ,  $180^\circ$ ,  $275^\circ$  and  $360^\circ$ .

Evaluate  $\sin 30^\circ \cos 60^\circ$ .

If  $\sec \theta = 13/12$  and  $\theta$  is acute, find  $\tan \theta$ .

Simplify  $\cos(180^\circ - \theta)$  and  $\tan(360^\circ - \theta)$ .

Find  $\cos 15^\circ$  exactly.

Write all six ratios when opposite=5, adjacent=12.

Prove numerically for  $\theta = 45^\circ$  that  $\sin^2 \theta + \cos^2 \theta = 1$ .

Find  $\sin 120^\circ$ ,  $\cos 240^\circ$  and  $\tan 315^\circ$ .

Find  $\sin 75^\circ$  exactly.

Evaluate  $\tan 75^\circ$  using the compound-angle formula.

## Practice answer guide

1. Acute, right, obtuse, straight, reflex, complete.

2. With hypotenuse 13:  $\sin = 5/13$ ,  $\cos = 12/13$ ,  $\tan = 5/12$ ,  $\operatorname{cosec} = 13/5$ ,  $\sec = 13/12$ ,  $\cot = 12/5$ .

3.  $1/4$ .

4.  $1/2 + 1/2 = 1$ .

5.  $\tan \theta = 5/12$ .

6.  $\sqrt{3}/2$ ,  $-1/2$ ,  $-1$ .

7.  $-\cos \theta$  and  $-\tan \theta$ .

8.  $(\sqrt{6} + \sqrt{2})/4$ .

9.  $(\sqrt{6} + \sqrt{2})/4$ .

10.  $2 + \sqrt{3}$ .

**Module II common mistakes:** wrong quadrant sign, mixing sine and cosine reduction formulae, and using the wrong denominator sign in  $\tan(A \pm B)$ .

## MODULE III

# Coordinate Geometry

Use coordinates to describe straight lines, their relative orientation, and circles.

11 Lecture Hours

4 Tutorial Hours

C03

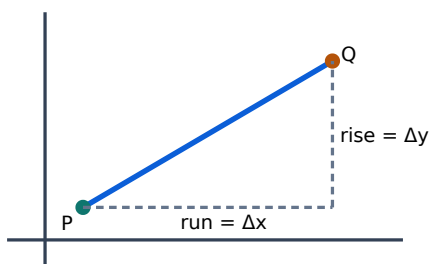
## 3.1 Slope and equation of a straight line—

The slope measures vertical change per unit horizontal change. Through two points  $P(x_1, y_1)$  and  $Q(x_2, y_2)$ :

$$m = (y_2 - y_1) / (x_2 - x_1), \quad x_2 \neq x_1$$

$m = \tan \theta$ , where  $\theta$  is the inclination from the positive x-axis.

Point-slope form:  $y - y_1 = m(x - x_1)$ .



$$\text{Slope } m = \text{rise/run} = \Delta y / \Delta x.$$

For a vertical line,  $x_2 - x_1 = 0$ , so the slope is not defined. Do not write  $m = 0$ .

### 3.2 Angle, parallel and perpendicular lines—

Angle between lines of slopes  $m_1$  and  $m_2$ :

$$\tan \theta = |(m_2 - m_1) / (1 + m_1 m_2)|.$$

**Parallel lines:**  $m_1 = m_2$ . **Perpendicular lines:**  $m_1 m_2 = -1$ , provided both slopes are finite.

- 1 Convert the given line to a form from which its slope can be read.
- 2 Use the same slope for a parallel line or the negative reciprocal for a perpendicular line.
- 3 Substitute the given point in  $y - y_1 = m(x - x_1)$ .
- 4 Simplify to the required form.

**Engineering mini-application:** line equations describe rails, cable routes, ramps and edges in a coordinate drawing. Parallel and perpendicular conditions help maintain alignment.

### 3.3 Equation of a circle—

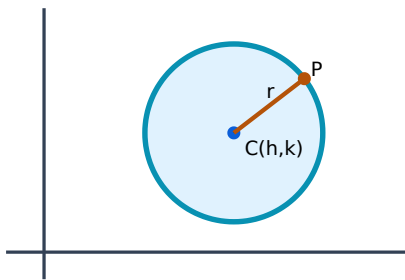
A circle is the set of points at a fixed distance  $r$  from centre  $C(h,k)$ .

$$(x-h)^2+(y-k)^2=r^2.$$

After expansion the general form is:

$$x^2+y^2+2gx+2fy+c=0,$$

$$\text{centre}=(-g,-f), \text{ radius}=\sqrt{g^2+f^2-c}.$$



Every point  $P$  on the circle satisfies  $CP=r$ .

In the general equation, compare  $2g$  and  $2f$  carefully. For example, coefficient  $-6x$  means  $2g=-6$ , hence  $g=-3$  and centre  $x$ -coordinate is  $-g=3$ .

### Worked examples



### Solved Example 3.1: Slope and point-slope form

**Question.** Find the equation of the line through (2,3) and (6,11).

**Rule.**  $m=(y_2-y_1)/(x_2-x_1)$ ;  $y-y_1=m(x-x_1)$ .

- 1  $m=(11-3)/(6-2)=8/4=2$ .
- 2 Using (2,3):  $y-3=2(x-2)$ .
- 3 Simplify:  $y-3=2x-4$ , so  $y=2x-1$ .

**Final answer:  $y=2x-1$ .**

**Check:** At  $x=6$ ,  $y=11$ .

### Solved Example 3.2: Angle between lines

**Question.** Find the acute angle between lines with slopes 1 and  $-1/3$ .

**Rule.**  $\tan\theta=|(m_2-m_1)/(1+m_1m_2)|$ .

- 1  $\tan\theta=|((-1/3)-1)/(1+1(-1/3))|$ .
- 2  $=|-4/3 \div 2/3|=2$ .
- 3 Therefore  $\theta=\tan^{-1}2$ , approximately  $63.435^\circ$ .

**Final answer:**

**$\theta=\tan^{-1}2\approx63.435^\circ$ .**

### Solved Example 3.3: Parallel and perpendicular lines

**Question.** Through (1,-2), find lines parallel and perpendicular to  $3x-2y+5=0$ .

**Rule.** Write the given line as  $y=(3/2)x+5/2$ , so  $m=3/2$ .

- 1 Parallel:  $y+2=(3/2)(x-1)$ . Multiply by 2:  $2y+4=3x-3$ .
- 2 Parallel line:  $3x-2y-7=0$ .
- 3 Perpendicular slope =  $-2/3$ .
- 4 Perpendicular:  $y+2=(-2/3)(x-1)$ . Multiply by 3 and simplify.

**Final answer: Parallel:**

**$3x-2y-7=0$ ; perpendicular:**

**$2x+3y+4=0$ .**

### Solved Example 3.4: Circle from centre and radius

**Question.** Form the equation of the circle with centre (3,-2) and radius 5.

**Rule.**  $(x-h)^2+(y-k)^2=r^2$ .

- 1 Substitute  $h=3$ ,  $k=-2$ ,  $r=5$ .
- 2  $(x-3)^2+(y+2)^2=25$ .
- 3 Expanded form:  
 $x^2+y^2-6x+4y-12=0$ .

**Final answer:  $(x-3)^2+(y+2)^2=25$ .**

### Solved Example 3.5: Centre and radius from general form

**Question.** Find the centre and radius of  $x^2+y^2-6x+8y-11=0$ .

**Rule.** Compare with  $x^2+y^2+2gx+2fy+c=0$ .

- 1  $2g=-6$ , so  $g=-3$ .  $2f=8$ , so  $f=4$ .  
 $c=-11$ .
- 2 Centre  $=(-g,-f)=(3,-4)$ .
- 3  $r=\sqrt{(g^2+f^2-c)}=\sqrt{(9+16+11)}=\sqrt{36}=6$ .

**Final answer: Centre (3, -4), radius 6.**

**Check:** Substitution gives the standard form  $(x-3)^2+(y+4)^2=36$ .

## Practice questions

Find the slope through  $(-1,4)$  and  $(3,-8)$ .

Find the equation through  $(2,-1)$  with slope 3.

Find the inclination of a line whose slope is  $1/\sqrt{3}$ .

Find the angle between lines with slopes 2 and  $1/3$ .

Test whether  $2x-3y+4=0$  and  $4x-6y-7=0$  are parallel.

Find the line through  $(4,1)$  parallel to  $x+2y-6=0$ .

Find the line through  $(-2,3)$  perpendicular to  $3x+y-4=0$ .

Form the circle with centre  $(-1,2)$  and radius 4.

Find centre and radius of  $x^2+y^2+4x-10y+13=0$ .

Convert  $(x-2)^2+(y+3)^2=49$  to general form.

## Practice answer guide

**1.**  $m = -3$ .

**6.**  $x + 2y - 6 = 0$ .

**2.**  $y = 3x - 7$ .

**7.**  $x - 3y + 11 = 0$ .

**3.**  $30^\circ$ .

**8.**  $(x+1)^2 + (y-2)^2 = 16$ .

**4.**  $45^\circ$ .

**9.** Centre  $(-2, 5)$ , radius 4.

**5.** Parallel; both have slope  $2/3$ .

**10.**  $x^2 + y^2 - 4x + 6y - 36 = 0$ .

**Module III common mistakes:** reversing only one difference in the slope formula, forgetting the absolute value in the angle formula, and reading centre coordinates with the wrong signs.

## MODULE IV

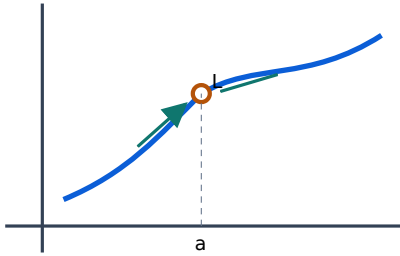
# Differential Calculus

Evaluate limits and use standard derivative rules to calculate first and second derivatives.

**12 Lecture Hours****3 Tutorial Hours****CO4**

## 4.1 Concept of a limit—

A limit describes the value approached by a function as  $x$  approaches a specified number. The function need not be evaluated exactly at that point.



As  $x$  approaches  $a$  from both sides,  $f(x)$  approaches  $L$ .

$x$  എന്ന value-ൽ എത്തേണ്ടതില്ല; അതിനോട് അടുത്തപ്പോൾ function ഏത് value-ലേക്ക് പോകുന്നു എന്നതാണ് limit.

## 4.2 Limits by substitution and factorisation—

Use direct substitution first. If it produces a finite value, stop. If it produces  $0/0$ , simplify by factorisation and then substitute.

- 1 Substitute the limiting value.
- 2 If the form is  $0/0$ , factorise numerator and denominator.
- 3 Cancel only a common factor, not separate terms.
- 4 Substitute again after simplification.

$$\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = na^{n-1}.$$

$$\lim_{x \rightarrow 0} \sin x/x = 1, \text{ where } x \text{ is measured in radians.}$$

**Do not cancel across addition or subtraction.** For example,  $x$  cannot be cancelled from  $(x+2)/x$ .

## 4.3 Differentiation and standard derivatives—

The derivative gives the instantaneous rate of change of  $y$  with respect to  $x$  and the slope of the curve at a point.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h.$$

| Function $y$             | $dy/dx$                          |
|--------------------------|----------------------------------|
| $k$                      | $0$                              |
| $x^n$                    | $nx^{n-1}$                       |
| $e^x$                    | $e^x$                            |
| $\log x$                 | $1/x$                            |
| $\sin x$                 | $\cos x$                         |
| $\cos x$                 | $-\sin x$                        |
| $\tan x$                 | $\sec^2 x$                       |
| $\operatorname{cosec} x$ | $-\operatorname{cosec} x \cot x$ |
| $\sec x$                 | $\sec x \tan x$                  |
| $\cot x$                 | $-\operatorname{cosec}^2 x$      |

These trigonometric derivatives assume  $x$  is in radians.

## 4.4 Linearity, product, quotient and second derivatives–

Linearity:  $d[af(x)+bg(x)]/dx = af'(x)+bg'(x)$ .

Product rule:  $d(uv)/dx = u(dv/dx)+v(du/dx)$ .

Quotient rule:  $d(u/v)/dx = [v(du/dx)-u(dv/dx)]/v^2$ .

Second derivative:  $d^2y/dx^2 = d/dx(dy/dx)$ .

| Question form                     | Choose                                   |
|-----------------------------------|------------------------------------------|
| Sum or difference of simple terms | Linearity and standard derivatives       |
| Two functions multiplied          | Product rule                             |
| One function divided by another   | Quotient rule                            |
| Second derivative requested       | Differentiate the first derivative again |

**Scope boundary:** chain rule, implicit differentiation, partial differentiation, maxima/minima and integration are not part of this Course 1002 syllabus.

**Engineering mini-application:** if displacement  $s(t)$  is known,  $ds/dt$  is velocity and  $d^2s/dt^2$  is acceleration. The same rate-of-change idea applies to temperature, voltage and production output.

## Worked examples

### Solved Example 4.1: Direct substitution and factorisation

**Question.** Evaluate (a)  $\lim_{x \rightarrow 2} (x^2 + 3x - 1)$ , (b)  $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$ .

**Rule.** Use substitution; for 0/0 factorise.

- 1 (a) Substitute  $x=2$ :  $4+6-1=9$ .
- 2 (b) Direct substitution gives 0/0.
- 3  $x^2-9=(x-3)(x+3)$ ; cancel  $x-3$  for  $x \neq 3$ .
- 4 Now substitute  $x=3$  in  $x+3$ .

**Final answer: (a) 9, (b) 6.**

### Solved Example 4.3: Standard trigonometric limit

**Question.** Evaluate  $\lim_{x \rightarrow 0} \sin 4x/x$ .

**Rule.** Create the standard form  $\sin u/u$ .

- 1 Write  $\sin 4x/x = 4[\sin 4x/(4x)]$ .
- 2 As  $x \rightarrow 0$ ,  $4x \rightarrow 0$  and  $\sin 4x/(4x) \rightarrow 1$ .
- 3 Therefore the limit is  $4 \cdot 1$ .

**Final answer: 4.**

**Check:** The angle is treated in radians.

### Solved Example 4.2: Limit of $x^n - a^n$ form

**Question.** Evaluate  $\lim_{x \rightarrow 2} (x^5 - 32)/(x - 2)$ .

**Rule.**  $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = na^{n-1}$ .

- 1 Here  $n=5$  and  $a=2$ .
- 2 Limit =  $5 \cdot 2^4 = 5 \cdot 16$ .

**Final answer: 80.**

### Solved Example 4.4: Linearity and standard results

**Question.** Differentiate  $y = 4x^5 - 3x^2 + 2e^x - 5\log x + 7$ .

**Rule.** Differentiate each term separately.

- 1  $d(4x^5)/dx = 20x^4$ ;  $d(-3x^2)/dx = -6x$ .
- 2  $d(2e^x)/dx = 2e^x$ ;  $d(-5\log x)/dx = -5/x$ .
- 3  $d(7)/dx = 0$ .

**Final answer:**  
 **$dy/dx = 20x^4 - 6x + 2e^x - 5/x$ .**

### Solved Example 4.5: Product and quotient rules

**Question.** Differentiate (a)  $y=x^2\sin x$ ,  
(b)  $y=(x^2+1)/x$ .

**Rule.** Product:  $u'v+uv'$ . Quotient:  
 $(vu'-uv')/v^2$ .

- 1 (a)  $u=x^2$ ,  $v=\sin x$ .  
 $y'=2x\sin x+x^2\cos x$ .
- 2 (b)  $u=x^2+1$ ,  $v=x$ .
- 3  $y'=[x(2x)-(x^2+1)(1)]/x^2$ .
- 4 Simplify:  $y'=(x^2-1)/x^2$ .

**Final answer:** (a)  $2x \sin x + x^2 \cos x$ ; (b)  $(x^2-1)/x^2$ .

### Solved Example 4.6: Second derivative

**Question.** If  $y=3x^4-2x^3+5x-7$ , find  $d^2y/dx^2$ .

**Rule.** Differentiate twice.

- 1 First derivative:  
 $dy/dx=12x^3-6x^2+5$ .
- 2 Differentiate again:  
 $d^2y/dx^2=36x^2-12x$ .

**Final answer:**  
 $d^2y/dx^2=36x^2-12x$ .

## Practice questions

Evaluate  $\lim_{x \rightarrow 1} (3x^2-2x+4)$ .

Evaluate  $\lim_{x \rightarrow 2} (x^3-8)/(x-2)$ .

Differentiate  $6x^7-4x^3+9$ .

Differentiate  $\sin x-2\cos x+\tan x$ .

Differentiate  $(\sin x)/x$  using the quotient rule.

Evaluate  $\lim_{x \rightarrow 4} (x^2-16)/(x-4)$ .

Evaluate  $\lim_{x \rightarrow 0} \sin 7x/x$ .

Differentiate  $e^x+3\log x$ .

Differentiate  $x^3\cos x$  using the product rule.

Find the second derivative of  $y=x^5-4x^3+2x$ .

## Practice answer guide

1. 5.

2. 8.  
Polytechnic Study Hub

3. 12.

4. 7.  
polypmna.dpdns.org



5.  $42x^6 - 12x^2$ .

8.  $3x^2 \cos x - x^3 \sin x$ .

6.  $e^x + 3/x$ .

9.  $(x \cos x - \sin x)/x^2$ .

7.  $\cos x + 2 \sin x + \sec^2 x$ .

10.  $d^2y/dx^2 = 20x^3 - 24x$ .

**Module IV common mistakes:** substituting before simplifying a 0/0 form, using degrees in  $\sin x/x$ , missing the minus sign for  $d(\cos x)/dx$ , and reversing terms in the quotient-rule numerator.

## LAST-MINUTE REVIEW

# Compact revision sheet

## Module I checklist

- Order and matrix types
- Transpose and compatible operations
- $2 \times 2$  and  $3 \times 3$  determinants
- Minor versus cofactor
- Adjoint and inverse condition
- Two- and three-variable Cramer's rule

## Module II checklist

- Six ratios and standard values
- Three Pythagorean relations
- ASTC signs
- Reduction formulae
- $\sin$ ,  $\cos$  and  $\tan$  of  $A \pm B$

## Module III checklist

- Slope from points and inclination
- Point-slope form

- Angle between lines
- Parallel/perpendicular conditions
- Standard and general circle forms

## Module IV checklist

- Direct substitution and factorisation
- $(x^n - a^n)/(x - a)$
- $\sin x/x$  in radians
- Standard derivative table
- Linearity, product, quotient
- Second derivative

**Five-minute error scan:** determinant signs, quadrant signs, centre signs, derivative signs, and the order of numerator terms in the quotient rule.

## EXAM PREPARATION

### Compact practice set

#### Part A style — 10 one-mark questions

1. State the order of a matrix having 3 rows and 2 columns.
2. Write the cofactor sign at position (2,3).
3. State the value of  $\cos 60^\circ$ .
4. In which quadrant is only tangent positive?
5. Write the slope of a line inclined at  $45^\circ$ .
6. State the condition for two finite-slope lines to be perpendicular.
7. Write the centre of  $x^2 + y^2 + 2gx + 2fy + c = 0$ .
8. Evaluate  $\lim_{x \rightarrow 0} \sin x/x$ .
9. Differentiate  $e^x$ .
10. Differentiate  $\cot x$ .

#### Part B style — 12 three-mark questions

1. Find  $3A - 2B$  for  $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ .
2. Evaluate  $\begin{vmatrix} 2 & 0 & 1 \\ 3 & -1 & 4 \\ 1 & 2 & 0 \end{vmatrix}$  by expansion along the first row.

3. Find the inverse of  $[[3,1],[5,2]]$ .

4. Find all ratios if  $\tan \theta = 5/12$  and  $\theta$  is acute.

5. Evaluate  $\sin 210^\circ$ ,  $\cos 300^\circ$  and  $\tan 135^\circ$ .

6. Find  $\sin 15^\circ$  using a compound-angle formula.

7. Find the equation through  $(-1,2)$  with slope  $-3$ .

8. Find the angle between lines of slopes  $2$  and  $-1$ .

9. Find centre and radius of  $x^2 + y^2 - 8x + 6y - 11 = 0$ .

10. Evaluate  $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$ .

11. Differentiate  $y = x^2 e^x$ .

12. Find the second derivative of  $y = 2x^5 - 3x^2 + 4$ .

## Part C style — 8 seven-mark questions

1. Solve a three-variable system by Cramer's rule and verify the result.

2. For a  $2 \times 2$  matrix, find determinant, minors, cofactors, adjoint and inverse; use the inverse or Cramer's rule to solve a related two-variable system.

3. Use standard values, ASTC and reduction formulae to evaluate a set of ratios of angles in different quadrants.

4. Derive exact values of  $\sin 75^\circ$ ,  $\cos 15^\circ$  and  $\tan 15^\circ$  using compound-angle formulae.

5. Find line equations parallel and perpendicular to a given line through specified points, and calculate the angle with another line.

6. Form a circle equation from centre and radius, then recover centre and radius from an expanded equation.

7. Evaluate limits using factorisation, the  $x^n - a^n$  result and the standard trigonometric limit.

8. Differentiate a polynomial/trigonometric product and a quotient, then calculate a second derivative.

**PRACTICE SUPPORT****Practice-set answer guide and engineering checks**

Use this section only after attempting the compact practice set. The purpose is to confirm the method and final result, not to replace written working.

**Answers to the 10 one-mark questions**

1.  $3 \times 2$ .

2. Negative sign.

3.  $1/2$ .

4. Quadrant III.

5.  $\tan 45^\circ = 1$ .

6.  $m_1 m_2 = -1$ .

7.  $(-g, -f)$ .

8. 1.

9.  $e^x$ .

10.  $-\operatorname{cosec}^2 x$ .

**Answers and method notes for the 12 three-mark questions**

1.  $3A - 2B = [[-1, 8], [-6, -11]]$ . Multiply first, then subtract corresponding entries.

2. Expansion gives  $2(-8) + 1(7) = -9$ . The middle term vanishes because its first-row entry is zero.

3. Determinant = 1, therefore  $A^{-1} = [[2, -1], [-5, 3]]$ .

4. Use a 5-12-13 triangle:  $\sin \theta = 5/13$ ,  $\cos \theta = 12/13$ ,  $\tan \theta = 5/12$  and take reciprocals for the other ratios.

5.  $\sin 210^\circ = -1/2$ ,  $\cos 300^\circ = 1/2$ ,  $\tan 135^\circ = -1$ .

6.  $\sin 15^\circ = \sin(45^\circ - 30^\circ) = (\sqrt{6} - \sqrt{2})/4$ .

7.  $y - 2 = -3(x + 1)$ , hence  $y = -3x - 1$ .

8.  $\tan \theta = 3$ , so the acute angle is  $\theta = \tan^{-1} 3 \approx 71.565^\circ$ .

9. Centre =  $(4, -3)$ , radius = 6.

10. Factor  $(x - 5)(x + 5)$ , cancel, then substitute: limit = 10.

$$11. \frac{dy}{dx} = e^x(x^2 + 2x).$$

$$12. \frac{dy}{dx} = 10x^4 - 6x; \frac{d^2y}{dx^2} = 40x^3 - 6.$$

## Method plan for the eight seven-mark questions

### 1. Three-variable Cramer

Write  $\Delta$  and all three replacement determinants clearly. A full answer must include  $x$ ,  $y$  and  $z$  and a quick substitution check.

### 2. Matrix and two-variable system

Show determinant, four minors, signed cofactors, adjoint and inverse. State the inverse condition before solving the equations.

### 3. Ratios in quadrants

For each angle, write its reference angle and quadrant. Award the sign using ASTC before inserting the standard value.

### 4. Compound angles

Choose a useful decomposition such as  $75^\circ = 45^\circ + 30^\circ$  or  $15^\circ = 45^\circ - 30^\circ$ . Write the complete formula before substitution.

### 5. Line relationships

Read the given slope, obtain the parallel or negative-reciprocal slope, use point-slope form, and simplify. Use the angle formula separately.

### 6. Circle conversion

Start with  $(x-h)^2 + (y-k)^2 = r^2$ . After expansion, compare  $2g$ ,  $2f$  and  $c$  and recover the same centre and radius.

## 7. Mixed limits

Classify each limit: direct substitution, factorisation,  $x^n - a^n$  form or  $\sin x/x$  form. Do not use a method that is not required.

## 8. Mixed derivatives

Mark  $u$  and  $v$  before applying product or quotient rule. Simplify the first derivative and differentiate again only when the second derivative is requested.

## Four engineering-oriented application checks

### Simultaneous electrical quantities

Two measured relations are  $2I_1 + I_2 = 11$  and  $I_1 - I_2 = 1$ . Cramer's rule gives  $\Delta = -3$ ,  $\Delta_1 = -12$ ,  $\Delta_2 = -9$ , hence  $I_1 = 4$  and  $I_2 = 3$ . Substitute both values to verify the readings.

### Inclined component check

A 100 N force acts at  $30^\circ$  to the horizontal. The horizontal component is  $100\cos 30^\circ = 50\sqrt{3}$  N and the vertical component is  $100\sin 30^\circ = 50$  N. This uses only the basic ratios of an acute angle.

### Coordinate route check

A cable route through  $(1,2)$  and  $(5,10)$  has slope 2 and equation  $y - 2 = 2(x - 1)$ , or  $y = 2x$ . A second route with slope  $-1/2$  through the same point is perpendicular.

### Rate-of-change check

Course 1002  
If a displacement model is  $s=2t^4-3t^2+5t$ , then velocity is  $ds/dt=8t^3-6t+5$  and acceleration is  $d^2s/dt^2=24t^2-6$ . The operation is standard first and second differentiation.

**Verification habit:** matrix answers can be checked by substitution, trigonometric answers by quadrant signs and approximate values, line/circle answers by point substitution, and derivatives by differentiating each term independently.

## LIKELY QUESTION FORMS

### Expected examination questions

#### Part A — 1 mark

- Definition, order or type of matrix
- One standard-angle value
- Parallel/perpendicular condition
- Centre of a circle
- Standard limit or derivative

#### Part B — 3 marks

- Matrix operation or  $2 \times 2$  inverse
- ASTC/reduction calculation
- Point-slope line or circle parameters
- Single limit or derivative rule

#### Part C — 7 marks

- Cramer's rule with full determinant work
- Compound-angle calculation set
- Parallel/perpendicular line plus circle work
- Mixed limits and derivatives

**Do not guess exact future questions.** These are syllabus-based question forms, not an official question-paper prediction.

## REVISION 2026 PATTERN

# Model Question Paper

## Diploma in Engineering and Technology — Semester I

### Course 1002 — Engineering Mathematics

Duration: 2.5 Hours | Maximum Marks: 60

#### Part A — Answer all questions. $8 \times 1 = 8$ marks

- |    |                                                                                   |          |
|----|-----------------------------------------------------------------------------------|----------|
| 1. | What is an identity matrix?                                                       | <b>1</b> |
| 2. | State the condition for a $2 \times 2$ matrix to have an inverse.                 | <b>1</b> |
| 3. | Write the value of $\sin 30^\circ$ .                                              | <b>1</b> |
| 4. | In which quadrant are sine ratios positive while cosine and tangent are negative? | <b>1</b> |
| 5. | Write the slope of the line joining $(x_1, y_1)$ and $(x_2, y_2)$ .               | <b>1</b> |
| 6. | Write the centre of $x^2 + y^2 + 2gx + 2fy + c = 0$ .                             | <b>1</b> |
| 7. | Evaluate $\lim_{x \rightarrow 0} \sin x/x$ .                                      | <b>1</b> |
| 8. | Differentiate $\cos x$ .                                                          | <b>1</b> |

#### Part B — Answer any two of the three questions from each module. $4 \times 6 = 24$ marks

##### Module I

- |       |                                                                                                                                         |          |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9(a). | For $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 0 & -2 \end{bmatrix}$ , find $A+B$ and $2A$ . | <b>3</b> |
| 9(b). | Evaluate $\begin{vmatrix} 2 & 1 & 0 \\ 3 & -1 & 2 \\ 1 & 4 & 5 \end{vmatrix}$ .                                                         | <b>3</b> |
| 9(c). | Find the inverse of $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ .                                                                    | <b>3</b> |

##### Module II

- |        |                                                                                       |          |
|--------|---------------------------------------------------------------------------------------|----------|
| 10(a). | If $\tan \theta = 8/15$ and $\theta$ is acute, find $\sin \theta$ and $\cos \theta$ . | <b>3</b> |
|--------|---------------------------------------------------------------------------------------|----------|



10(b). Evaluate  $\sin 150^\circ$ ,  $\cos 240^\circ$  and  $\tan 315^\circ$ . 3

10(c). Find  $\cos 75^\circ$  using a compound-angle formula. 3

### Module III

11(a). Find the equation of the line through  $(2, -1)$  with slope 4. 3

11(b). Find the acute angle between lines whose slopes are  $1/2$  and  $-1$ . 3

11(c). Find the centre and radius of  $x^2 + y^2 - 4x + 6y - 12 = 0$ . 3

### Module IV

12(a). Evaluate  $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$ . 3

12(b). Differentiate  $y = 3x^4 - 5x^2 + 2e^x$ . 3

12(c). Differentiate  $y = x^2 \sin x$ . 3

## Part C — Answer either (a) or (b) from each module. $4 \times 7 = 28$ marks

### Module I

13(a). Solve  $2x + y = 7$  and  $3x - 2y = 0$  using Cramer's rule. Also find the inverse of the coefficient matrix. 7

13(b). Solve  $x + y + z = 6$ ,  $2x - y + z = 3$ ,  $x + 2y - z = 2$  by Cramer's rule. 7

### Module II

14(a). Using compound-angle formulae, find exact values of  $\sin 75^\circ$ ,  $\cos 15^\circ$  and  $\tan 15^\circ$ . 7

14(b). Evaluate  $\sin 120^\circ$ ,  $\cos 210^\circ$ ,  $\tan 300^\circ$ ,  $\sin(45^\circ + 30^\circ)$  and  $\cos(60^\circ - 45^\circ)$ , showing signs and formulae. 7

### Module III

15(a). Find equations of the lines through  $(2, 3)$  parallel and perpendicular to  $2x - 3y + 5 = 0$ . Also find the angle between the two lines. 7

15(b). Form the circle with centre  $(-2, 3)$ , radius 5. Expand it and determine centre and radius again from the general form. 7

### Module IV

16(a). Evaluate  $\lim_{x \rightarrow 2} (x^4 - 16)/(x - 2)$  and  $\lim_{x \rightarrow 0} \sin 5x/x$ . Then differentiate 7

$y = x^3 \cos x$ .

$$y=2x^4-3x^3+x.$$

**Total: 8 + 24 + 28 = 60 marks.**

## SELF-CHECK

# Answer key and model answers

## Part A direct answers

**1.** A square matrix with 1 on the principal diagonal and 0 elsewhere.

**2.** Determinant must be non-zero.

**3.**  $1/2$ .

**4.** Quadrant II.

**5.**  $(y_2-y_1)/(x_2-x_1)$ .

**6.**  $(-g, -f)$ .

**7.** 1.

**8.**  $-\sin x$ .

## Part B concise method outlines

**9(a)** Add corresponding entries:  $A+B = [[3,4],[3,2]]$ . Multiply each A entry by 2:  $2A = [[4,-2],[6,8]]$ .

**9(b)** Expand along row 1:  
 $2[(-1)5-2(4)]$   
 $-1[3(5)-2(1)] = 2(-13)-13 = -39$ .

**9(c)**  $\Delta = 3(1)-1(2) = 1$ .  $A^{-1} = [[1,-1], [-2,3]]$ .

**10(a)** Triangle sides 8,15,17.  
 $\sin\theta = 8/17$ ,  $\cos\theta = 15/17$ .

**10(b)**  $\sin 150^\circ = 1/2$ ;  $\cos 240^\circ = -1/2$ ;  
 $\tan 315^\circ = -1$ .

**10(c)**  $\cos 75^\circ = \cos(45^\circ + 30^\circ) = (\sqrt{6}-\sqrt{2})/4$ .

**11(a)**  $y+1=4(x-2)$ , hence  $y=4x-9$ .

**11(b)**  $\tan\theta = |[-1-1/2]/[1-1/2]| = 3$ ,  
hence  $\theta = \tan^{-1}3$ .

**11(c)**  $2g = -4 \Rightarrow g = -2$ ;  $2f = 6 \Rightarrow f = 3$ ;  
 $c = -12$ . Centre =  $(2, -3)$ , radius = 5.

**12(a)** Factor to  $x+3$ , so limit=6.

**12(c)**  $dy/dx=2x \sin x+x^2 \cos x$ .

**12(b)**  $dy/dx=12x^3-10x+2e^x$ .

## Selected complete Part C solutions

### 13(b) — Three-variable Cramer's rule

- 1 Coefficient determinant  $\Delta=|1 \ 1 \ 1; 2 \ -1 \ 1; 1 \ 2 \ -1|=7$ .
- 2  $\Delta x=|6 \ 1 \ 1; 3 \ -1 \ 1; 2 \ 2 \ -1|=7$ .
- 3  $\Delta y=|1 \ 6 \ 1; 2 \ 3 \ 1; 1 \ 2 \ -1|=14$ .
- 4  $\Delta z=|1 \ 1 \ 6; 2 \ -1 \ 3; 1 \ 2 \ 2|=21$ .
- 5  $x=1, y=2, z=3$ ; substitution verifies all equations.

**Answer:  $(x,y,z)=(1,2,3)$ .**

### 14(a) — Compound angles

- 1  $\sin 75^\circ = \sin(45^\circ + 30^\circ) = (\sqrt{6} + \sqrt{2})/4$ .
- 2  $\cos 15^\circ = \cos(45^\circ - 30^\circ) = (\sqrt{6} + \sqrt{2})/4$ .
- 3  $\tan 15^\circ = \tan(45^\circ - 30^\circ) = (1 - 1/\sqrt{3})/(1 + 1/\sqrt{3}) = 2 - \sqrt{3}$ .

**Answers:  $\sin 75^\circ = (\sqrt{6} + \sqrt{2})/4$ ,  $\cos 15^\circ = (\sqrt{6} + \sqrt{2})/4$ ,  $\tan 15^\circ = 2 - \sqrt{3}$ .**

### 15(b) — Circle equation and recovery

- 1 Centre  $(-2,3)$ ,  $r=5$  gives  $(x+2)^2 + (y-3)^2 = 25$ .
- 2 Expand:  $x^2 + 4x + 4 + y^2 - 6y + 9 = 25$ .
- 3 General form:  $x^2 + y^2 + 4x - 6y - 12 = 0$ .
- 4  $2g=4 \Rightarrow g=2$ ;  $2f=-6 \Rightarrow f=-3$ ;  $c=-12$ .
- 5 Centre  $=(-2,3)$ ;  $r=\sqrt{4+9+12}=5$ .

**Answer:  $(x+2)^2 + (y-3)^2 = 25$ .**

## 16(a) — Limits and product derivative

- 1  $\lim_{x \rightarrow 2} (x^4 - 16)/(x - 2) = 4 \cdot 2^3 = 32.$
- 2  $\lim_{x \rightarrow 0} \sin 5x/x = 5[\sin 5x/(5x)] = 5.$
- 3 For  $y = x^3 \cos x$ , use product rule.
- 4  $dy/dx = 3x^2 \cos x - x^3 \sin x.$

**Answers: 32, 5, and  $3x^2 \cos x - x^3 \sin x$ .**

## INDEPENDENT STUDY

### Self-learning activities

The official plan totals **60 self-learning hours**. Use the checklist below to record completion without reproducing the full syllabus table.

#### Module I

- Matrix operations
- Cramer's rule
- Minors, cofactors, adjoint and inverse
- Engineering applications

#### Module II

- Ratios of acute angles
- Ratios in different quadrants
- Compound-angle problems
- Engineering applications

#### Module III

- Slope and line equations
- Parallel/perpendicular lines
- Circle centre and radius
- Engineering applications

## Module IV

- Limits
- Standard derivatives
- Product, quotient and second derivative
- Engineering applications

☐ Completed one worked example from every Module I subtopic.

☐ Prepared the standard-angle and reduction tables from memory.

☐ Drew and labelled one straight-line and one circle diagram.

☐ Reproduced the derivative table and rule-selection guide.

### LEARNING RESOURCES

## References

### Official textbook

**Fundamentals of Engineering Mathematics** — SITTTTR Kalamassery.

### Selected syllabus references

- *Schaum's Outline of Linear Algebra* — Seymour Lipschutz and Marc Lipson, McGraw Hill.
- *Schaum's Outline of Trigonometry* — Robert E. Moyer, McGraw Hill.
- *Schaum's Outline of Geometry* — Barnett Rich and Christopher Thomas, McGraw Hill.
- *Schaum's Outline of Calculus* — Frank Ayres and Elliott Mendelson, McGraw Hill.
- *Advanced Engineering Mathematics*, 10th edition — Erwin Kreyszig, Wiley & Sons.
- *A Textbook of Engineering Mathematics* — N. P. Bali and Manish Goyal, University Science Press.
- *Advanced Engineering Mathematics* — B. S. Grewal, Mercury Learning & Information.

**Official web resources:** NIL. No invented web resource list has been added.